MILITARY SPECIFICATION

CLAMP, BONE AND PLATE HOLDING,
Lowman's Modified

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bone and plate holding clamps of the modified Lowman's design, for use in the practice of orthopedic surgery.

1.2 Classification. The clamps covered by this specification shall be of the following sizes as specified (see 6.2).

Sizes	Length (in inches)
1	4-3/4
2	6-3/4
3	8-1/2

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

1830	CLAMP, BONE AND PLATE HOLDING, Lowman's Modified.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The clamps shall be in accordance with the applicable requirements of GG-I-526, and the following option exercises under that specification.

3.2 Material and hardness. Shall be as indicated in Table I:

Table I. Materials and Hardness

Component	Class	Type	Hardness
Upper & lower members of clamp	4	420	C40-C45
Screw and guide pin	4	416	B75-B85

The hardness requirements specified shall be applicable for the involved components after proper stress relieving.

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3.3 Style, design and dimensions. The bone and plate holding clamp shall be of the Lowman modified design, consisting essentially of two members with two distal and one proximal fingers, so arranged as to approach under screw pressure and exert pressure between the lower or distal and upper points. Shall be in strict accordance with Defense Personnel Support Center Drawing No. 1830.

3.4 Finish. The finish shall be mirror, mirror and satin, or satin.

3.5 Identification marking. In addition to GG-I-526 requirements, clamps of foreign manufacture shall also be marked "stainless".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, ndividual test results, and the specification requirements.

4.2 Sampling.

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4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table II. The unit of product for examination shall be on knife.

Table II. Sampling for Examination

Type of defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. Unit of product for test purposes shall be one clamp.

Table III. Sampling for Tests

Component (unit of product)	Characteristic	Req.	Test procedures	Insp. level	AQL (percent defective)
1-knife	Composition analysis	3.2 & GG-I-526	GG-I-526	**	*
	Hardness	3.2 & GG-I-526	GG-I-526	S-2	1.0
	Dimensions	3.3	- - - -	S-2	4.0
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Copper sulfate	GG-I-526	GG-I-526	S-2	1.0

* Acceptance number shall be zero for all sample sizes.
** Sufficient amount from each lot of each metal used in the construction of the clamp shall be selected for composition analysis.

4.3 Examination. The clamp shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any clamp in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

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Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Instrument not constructed as specified.
102	Style and design not as specified.
103	Instrument not free of sharp edges, rough areas, burrs, cracks, grind marks or overlaps.
104	Serrations not as specified.
Minor	
201	Identification marking not present, correct, complete, or legible.
202	Finish not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements in accordance with GG-I-526 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One clamp, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Preservative shall not be applied.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1C-2.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. Commercial colors are acceptable on intermediate cartons and closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3, PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-501, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bone and plate holding clamps are intended for use in the practice of orthopedic surgery.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification is intended to cover the following items listed in the Federal Supply catalog:

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National Stock Number	Item Identification	Size No.
6515-00-319-9100	CLAMP, BONE AND PLATE HOLDING, Lowman's Modified, 4-3/4 inch.	1
6515-00-319-9200	CLAMP, BONE AND PLATE HOLDING, Lowman's Modified, 6-3/4 inch.	2
6515-00-319-9300	CLAMP, BONE AND PLATE HOLDING, Lowman's Modified, 8-1/2 inch.	3

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1762

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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